



St. MARTIN'S ENGINEERING COLLEGE

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ASSIGNMENT ANSWER BOOKLET

Hall Ticket No. :

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Name of the Student: A. Deekshitha.

Course : B-Tech.

Branch : CSE

Year : I/II/III/IV

Sem : I/II

Assignment : I/II

Subject Name : Computer Graphics.

Last Date of Submission: 31/8/26

MARKS AWARDED

Q.No.	a	b	Total
1			
2			
3			
4			
5			
Total			

Actual Date of Submission : 27/8/26

A. Deekshitha
Signature of the Student

Signature of the Evaluator

Start Writing from Here

1. Explain Bresenham's Line Drawing Algorithm with an example.

$$y = mx + c$$

$$y = m(x_k + 1) + c$$

$$d_1 = y - y_k = m(x_k + 1) + c - y_k$$

$$d_2 = y_{k+1} - y = y_{k+1} - m(x_k + 1) - c$$

$$\begin{aligned} d_1 - d_2 &= (m(x_k + 1) + c - y_k) - (y_{k+1} - m(x_k + 1) - c) \\ &= (m(x_k + 1) + c - y_k - y_{k+1} + m(x_k + 1) + c) \\ &= 2m(x_k + 1) - 2y_k + 2c - 1 \\ &= 2 \left(\frac{\Delta y}{\Delta x} \right) (x_k + 1) - 2y_k + 2c - 1 \end{aligned}$$

Multiply both sides with ' Δx '

$$\begin{aligned} \Delta x(d_1 - d_2) &= \Delta x \left[2 \frac{\Delta y}{\Delta x} (x_k + 1) - 2y_k + 2c - 1 \right] \\ &= 2\Delta y(x_k + 1) - 2\Delta x y_k + 2\Delta x c - \Delta x \end{aligned}$$



$$= 2\Delta y x_k + 2\Delta y - 2\Delta x y_k + 2\Delta x c - \Delta x$$

Group the three terms into a constant.

$$C = 2\Delta y + 2\Delta x c - \Delta x$$

$$\therefore P_k = 2\Delta y x_k - 2\Delta x y_k + C \quad \text{--- (1)}$$

Successive Parameter Decision.

$$P_{k+1} = 2\Delta y x_{k+1} - 2\Delta x y_{k+1} + C \quad \text{--- (2)}$$

$$\begin{aligned} P_{k+1} - P_k &= 2\Delta y x_{k+1} - 2\Delta x y_{k+1} + C - 2\Delta y x_k + 2\Delta x y_k - C \\ &= 2\Delta y (x_{k+1} - x_k) + 2\Delta x (y_k - y_{k+1}) \end{aligned}$$

$$\text{If } P_k < 0 \Rightarrow (x_{k+1}, y_{k+1}) = (x_k, y_k)$$

$$P_{k+1} = P_k + 2\Delta y$$

$$\text{If } P_k > 0 \Rightarrow (x_{k+1}, y_{k+1}) = (x_{k+1}, y_{k+1})$$

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x$$

Initial P value

$$P_k = 2\Delta y x_k - 2\Delta x y_k + 2\Delta y + 2\Delta x c - \Delta x$$

$$\begin{aligned} K=0 \quad P_0 &= 2\Delta y x_0 - 2\Delta x y_0 + 2\Delta y + 2\Delta x c - \Delta x \\ &= 2\Delta y x_0 - 2\Delta x y_0 + 2\Delta y + 2\Delta x \left[y_0 - \frac{\Delta y}{\Delta x} x_0 \right] - \Delta x \\ &= 2\Delta y x_0 - 2\Delta x y_0 + 2\Delta y + 2\Delta x y_0 - 2\Delta y x_0 - \Delta x \end{aligned}$$

$$\therefore P_0 = 2\Delta y - \Delta x$$

Example; (2,2), (6,5) are the points.

Start point: $(x_0, y_0) = (2, 2)$ and Ending point: $(x_1, y_1) = (6, 5)$

Step-1: Find the slope

$$m = \frac{y_1 - y_0}{x_1 - x_0} = \frac{5-2}{6-2} = \frac{3}{4} = 0.75$$

Since $m < 1$

Step-2: calculate initial decision parameter

$$P_0 = 2\Delta y - \Delta x$$

$$P_0 = 2(3) - 4 = 2$$

So, $P_0 > 0$. Therefore $x_{k+1} = x_k + 1, y_{k+1} = y_k$

Starting from (2,2): $(2+1, 2) = (3, 2)$

$m < 1$	$m \geq 1$
$P < 0$	$P < 0$
$x_{k+1} = x_k + 1$	$x_{k+1} = x_k + 1$
$y_{k+1} = y_k$	$y_{k+1} = y_k + 1$
$P_{k+1} = P_k + 2\Delta y$	$P_{k+1} = P_k + 2\Delta x$
$P \geq 0$	$P \geq 0$
$x_{k+1} = x_k + 1$	$x_{k+1} = x_k + 1$
$y_{k+1} = y_k + 1$	$y_{k+1} = y_k + 1$
$P_{k+1} = P_k + 2\Delta y - 2\Delta x$	$P_{k+1} = P_k + 2\Delta y - 2\Delta x$



Step-3: Find P_1

When $P_k > 0$, then $P_{k+1} = P_k + 2\Delta y - 2\Delta x$

$$P_{0+1} = P_0 + 2\Delta y - 2\Delta x$$

$$= 2 + 6 - 8$$

$$P_1 = 0$$

As $P_k \geq 0$ now $x_{k+1} = x_k + 1$ & $y_{k+1} = y_k + 1$

So from $(3, 3)$: $x_2 = 3 + 1 = 4$, $y_2 = 3 + 1 = 4$

Therefore : $(4, 4)$.

Step-4: Find P_2

$$P_{1+1} = P_1 + 2(3) - 2(4)$$

$$= 0 + 6 - 8$$

$$P_2 = -2$$

Now $P_2 < 0$. So, $x_{k+1} = x_k + 1$ & $y_{k+1} = y_k$

from $(4, 4)$: $x_3 = 4 + 1 = 5$, $y_3 = 4$

Therefore : $(5, 4)$.

Step-5: Find P_3

For $P_k < 0$: $P_{k+1} = P_k + 2\Delta y$

Therefore : $P_{2+1} = P_2 + 2(3)$

$$= -2 + 6$$

$$P_3 = 4$$

Since $P_3 > 0$.

From $(5, 4)$: $x_4 = 5 + 1 = 6$, $y_4 = 4 + 1 = 5$.

Therefore : $(6, 5)$.

Therefore, using Bresenham's line drawing algorithm, the line from $(2, 2)$ to $(6, 5)$ is drawn.

Describe the Scan-line Polygon Filling Algorithm with illustration

Scan-Line Algorithm

1. Draw the Polygon and Sort y_{min} to y_{max} .

First draw the polygon using the given vertices.

2. Construct the Edge table

For every edge:



- Ignore horizontal edges

- Find Y_{max} $X(Y_{min})$ $1/m$

- Store these values in edge table.

3. Insert the edges into Active Edge table and find the intersecting points.

4. Remove the completed edges from Active edge table.

5. After filling the one scan line, move to the next scan line by updating the intersection X -value using:

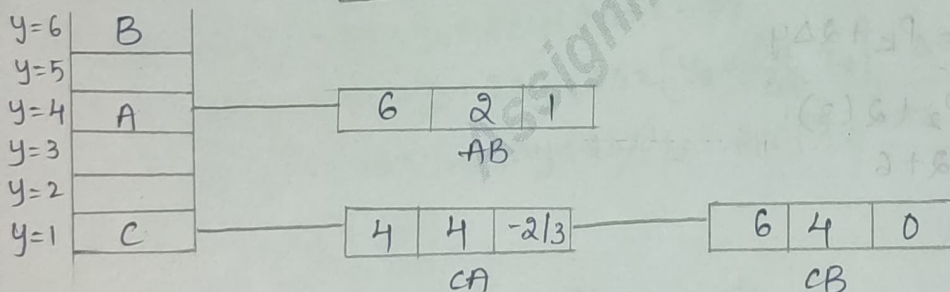
$$X_{new} = X_{old} + \frac{1}{m}$$

6. When there are no more edges in both the Edge Table and Active Edge Table, the entire polygon has been filled.

Ex: $A(2,4)$, $B(4,6)$, $C(4,1)$

Sorting Y_{min} to Y_{max} : $C(4,1)$, $A(2,4)$, $B(4,6)$.

Edge table:



Edge Table:

Edge	Y_{max}	$X(Y_{min})$	$1/m$
CA	4	4	-2/3
CB	6	4	0
AB	6	2	1

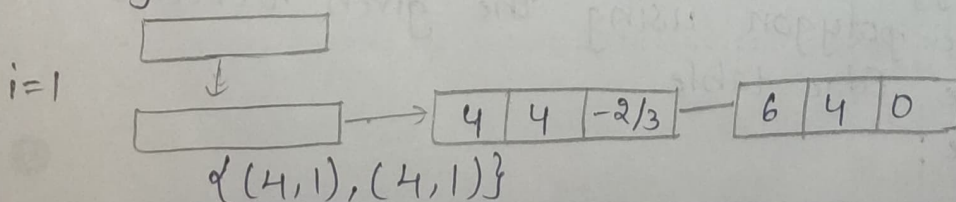
CA $\rightarrow C(4,1), A(2,4)$; $m = \frac{4-1}{2-4} = -\frac{3}{2}$; $\frac{1}{m} = -\frac{2}{3}$

CB $\rightarrow C(4,1), B(4,6)$; $m = \frac{6-1}{4-4} = \frac{5}{0}$; $\frac{1}{m} = 0$

AB $\rightarrow A(2,4), B(4,6)$; $m = \frac{6-4}{4-2} = \frac{2}{2} = 1$; $\frac{1}{m} = 1$

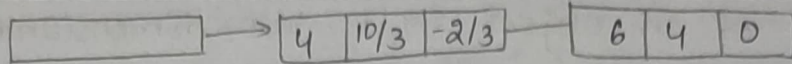
Active edge table:

It begins with null node and ends with null node.





$i=2$

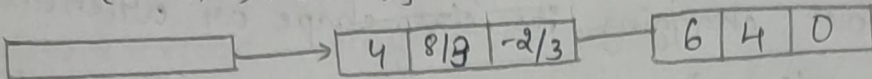


$$x_{k+1} = 4 - 2/3 = \frac{10}{3} = 3.4$$

$$x_{k+1} = x_k + \frac{1}{m}$$

$\{(3.4, 2), (4, 2)\}$

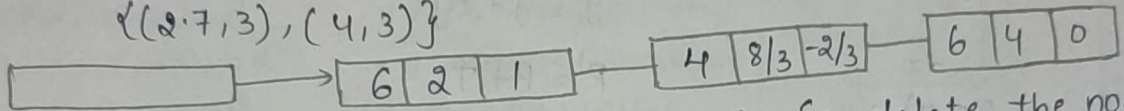
$i=3$



$$x_{k+1} = \frac{10}{3} - \frac{2}{3} = \frac{8}{3} = 2.67$$

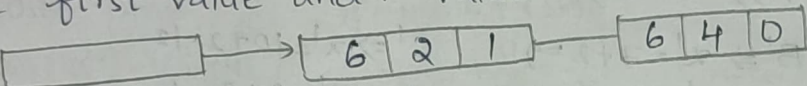
$\{(2.7, 3), (4, 3)\}$

$i=4$



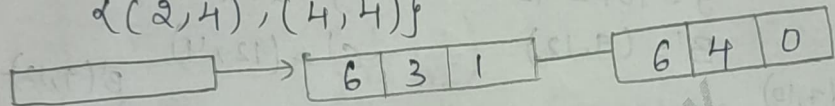
As with 3 nodes we cannot make a pair. So, delete the node whose first value and i value are same.

$i=4$



$\{(2, 4), (4, 4)\}$

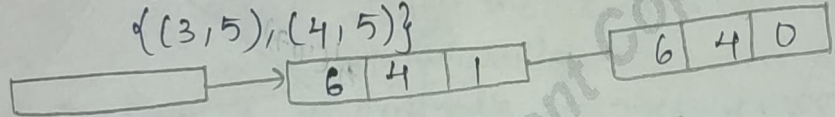
$i=5$



$$x_{k+1} = 2 + 1 = 3$$

$\{(3, 5), (4, 5)\}$

$i=6$

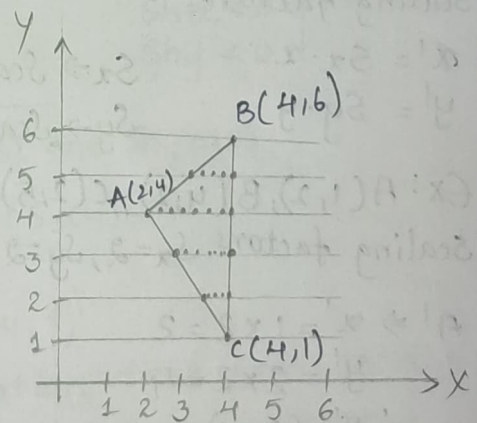


$$x_{k+1} = 3 + 1 = 4$$

$\{(4, 6), (4, 6)\}$

Active Edge Table.

Edge	Active Edges	X intersection	Pixels Filled
1	CA, CB	4, 4	None
2	CA, CB	3.4, 4	(4, 2)
3	CA, CB	2.67, 4	(3, 3), (4, 3)
4	AB, CB	2, 4	(2, 4), (3, 4), (4, 4)
5	AB, CB	3, 4	(3, 5), (4, 5)
6	None	—	None.



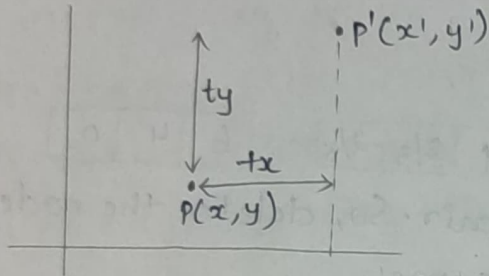
3 Explain 2D Transformation with a suitable example.
It is an operation that changes the position, size, orientation,



shape of an object in a two-dimensional coordinate system.

Types of 2D Transformation

1. Translation: It moves an object from one position to another without changing its size, shape or orientation.



$$x' = x + tx$$

$$y' = y + ty$$

$$P' = P + T$$

Ex: $A(2, 5)$, $B(7, 10)$, $C(10, 2)$ and $tx=2$, $ty=2$

$$A' \Rightarrow x' = 2 + 2 = 4$$

$$y' = 5 + 2 = 7$$

$$A' = (4, 7)$$

$$B' \Rightarrow x' = 7 + 2 = 9$$

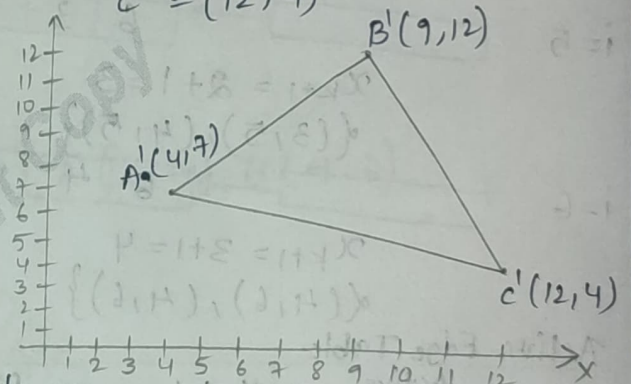
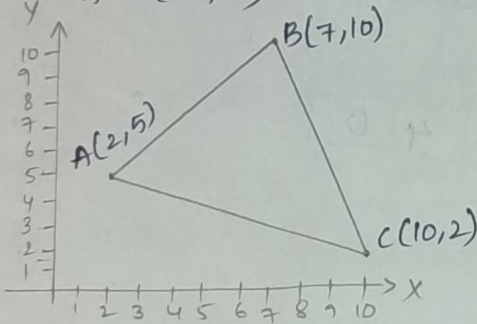
$$y' = 10 + 2 = 12$$

$$B' = (9, 12)$$

$$C' \Rightarrow x' = 10 + 2 = 12$$

$$y' = 2 + 2 = 4$$

$$C' = (12, 4)$$



2. Scaling: It can make an object larger or smaller without changing its basic shape. Co-ordinates of every point are multiplied by scaling factor.

$$x' = Sx \cdot x$$

$$y' = Sy \cdot y$$

$Sx \rightarrow$ Scaling factor along X-axis.

$Sy \rightarrow$ Scaling factor along Y-axis.

Ex: $A(1, 2)$, $B(4, 2)$, $C(2, 5)$

Scaling factors $Sx=2$, $Sy=2$

$$A' \Rightarrow x' = 1 \times 2 = 2$$

$$y' = 2 \times 2 = 4$$

$$\therefore A' = (2, 4)$$

$$B' \Rightarrow x' = 4 \times 2 = 8$$

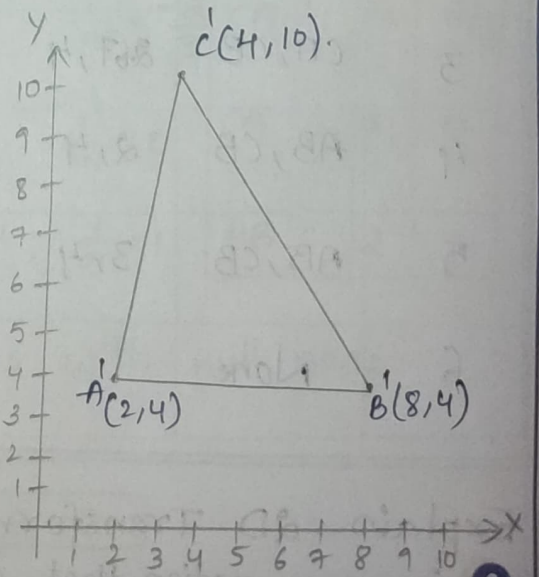
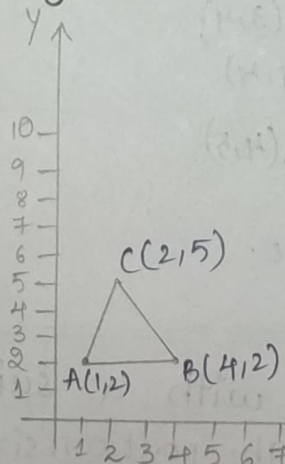
$$y' = 2 \times 2 = 4$$

$$\therefore B' = (8, 4)$$

$$C' \Rightarrow x' = 2 \times 2 = 4$$

$$y' = 5 \times 2 = 10$$

$$\therefore C' = (4, 10)$$





3. Rotation: A geometric transformation that turns an object through a specified angle about a fixed point.

During rotation the shape and size of object remains unchanged only orientation of object changes.

$$\cos \alpha = \frac{x}{r} \Rightarrow x = r \cos \alpha \quad \sin \alpha = \frac{y}{r} \Rightarrow y = r \sin \alpha$$

$$\cos(\theta + \alpha) = \frac{x'}{r} \Rightarrow x' = r \cos(\theta + \alpha) \quad \sin(\theta + \alpha) = \frac{y'}{r} \Rightarrow y' = r \sin(\theta + \alpha)$$

$$= r(\cos \theta \cos \alpha - \sin \theta \sin \alpha) \quad = r(\sin \theta \cos \alpha + \cos \theta \sin \alpha)$$

$$= r \cos \theta \cos \alpha - r \sin \theta \sin \alpha \quad = r \sin \theta \cos \alpha + r \cos \theta \sin \alpha$$

$$x' = x \cos \theta - y \sin \theta \quad y' = x \sin \theta + y \cos \theta$$

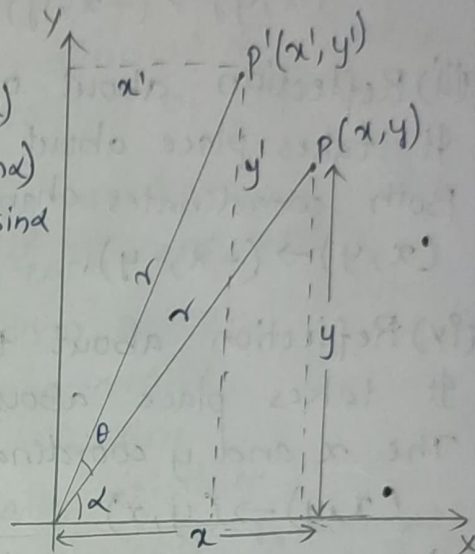
ex: $P(4, 3)$ and angle $(\theta) = 45^\circ$

$$x' = x \cos \theta - y \sin \theta \quad y' = x \sin \theta + y \cos \theta$$

$$= x \cos(45^\circ) - y \sin(45^\circ) \quad = \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{7}{\sqrt{2}}$$

$$= \frac{4}{\sqrt{2}} - \frac{3}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$P' = \left(\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}} \right)$$

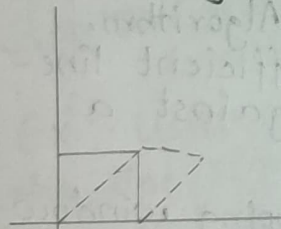


4. Shearing: A geometric transformation that slants the shape of an object along the x-axis or y-axis without changing its area significantly.

X-shearing: Only x-coordinate changes, y-coordinate remains same.

Y-shearing: Only y-coordinate changes, x-coordinate remains unchanged.

X-shearing



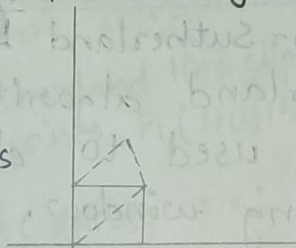
$$x' = x + Sh_x \cdot y$$

$$y' = y$$

$$Sh_x > 0 \Rightarrow +ve \text{ axis}$$

$$Sh_y > 0$$

Y-shearing.



$$x' = x$$

$$y' = y + Sh_y \cdot x$$

$$Sh_x < 0 \Rightarrow -ve \text{ axis}$$

$$Sh_y < 0$$

ex: $P = (4, 3)$ and $Sh_x = 2$

$$x' = 4 + 2(3) = 10$$

$$y' = 3$$

$$\therefore P' = (10, 3)$$

ex: $P = (4, 3)$ and $Sh_y = 1.5$

$$y' = 3 + (1.5 \times 4) = 9$$

$$x' = 4$$

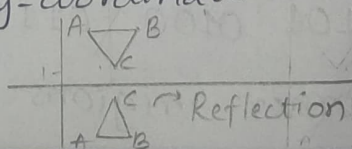
$$\therefore P' = (4, 9)$$

5. Reflection: A geometric transformation that produces the mirror image of an object with respect to a reference line called axis of reflection or mirror line.

Types:

(i) Reflection about X-axis: Object is reflected about x-axis. The x-coordinate remains unchanged, y-coordinate changes its sign.

$$x' = x, y' = -y$$



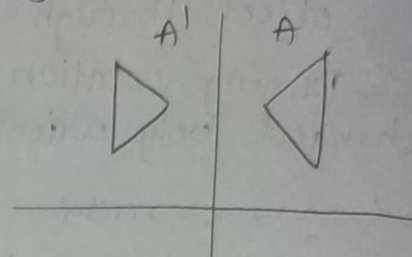
Reflection.



(ii) Reflection about Y-axis: The object is reflected about Y-axis.

→ The y-coordinate remains same, x-coordinate changes its sign.

$$(x, y) \rightarrow (-x, y)$$

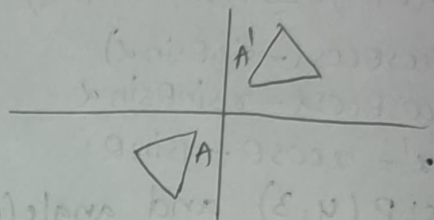


(iii) Reflection about origin:

It takes place about the origin (0,0).

Both coordinates change their sign.

$$(x, y) \rightarrow (-x, -y)$$

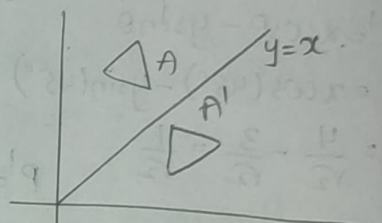


(iv) Reflection about the line $y = x$:

It takes place about the line $y = x$.

The x and y coordinates are interchanged.

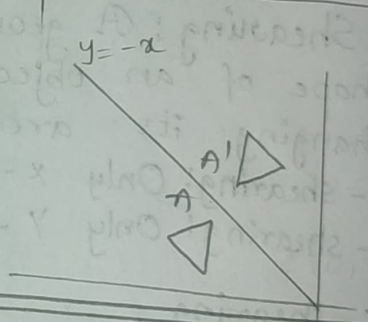
$$(x, y) \rightarrow (y, x)$$



(v) Reflection about the line $y = -x$:

The coordinates are interchanged and both signs are changed.

$$(x, y) \rightarrow (-y, -x)$$



H Describe the Cohen-Sutherland Line clipping Algorithm.

* The Cohen-Sutherland algorithm is an efficient line-clipping algorithm used to clip a line against a rectangular clipping window.

* It divides the 2D plane around the clipping window into 9 regions and assigns a 4-bit region code to each endpoint of the line.

Region code:

1001	1000	1010
0001	0000	0010
0101	0100	0110

0000 → Inside the window.

0001 → Left

0010 → Right

0100 → Bottom

1000 → Top

The corner regions contain combination of these bits.



Three possibilities for the line.

- * Line can be completely inside the window. It should be accepted.
- * Line can be completely outside the window. The line will be completely removed from the region.
- * Line can be partially inside the window. We find intersection point and draw only portion of line that is inside region.

Steps of Algorithm:

- Step 1: Consider the rectangular clipping window.
- Step 2: Assign a 4-bit region code to each endpoint.
- Step 3: If both endpoints have code 0000, accept the line.
- Step 4: Perform bitwise AND of the two region codes.
- Step 5: If C_1 and $C_2 \neq 0000$, then reject the line.
- Step 6: Otherwise, the line is partially visible.
- Step 7: Select an endpoint whose region code is non-zero.
- Step 8: Determine which boundary the endpoint lies outside.
- Step 9: Calculate the intersection of line with boundary.
- Step 10: Replace the outside endpoint with intersection point.

is either accepted or rejected.

Ex :- Using the Cohen-Sutherland line clipping algorithm, clip the line segment joining the points $P_1(40, 15)$ and $P_2(75, 45)$.

From the given diagram, the values are:

$$A = (40, 15)$$

$$B = (75, 45)$$

$$x_{\min} = 50, x_{\max} = 80$$

$$y_{\min} = 10, y_{\max} = 40$$

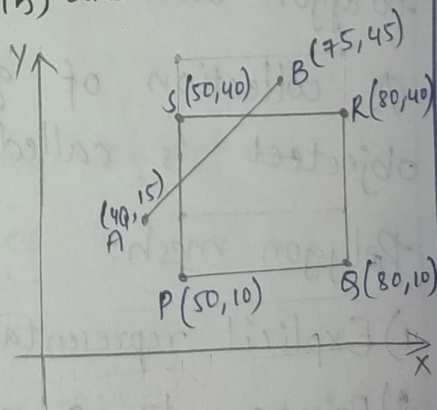
Region codes

$P_1(40, 15)$ is to the left. $P_1 = 0001$

$P_2(75, 45)$ is to above. $P_2 = 1000$

AND: $0001 \text{ AND } 1000 = 0000$.

The line is partially visible we need intersecting points.





$$\text{Slope (m)} = \frac{45-15}{75-40} = \frac{30}{35} = \frac{6}{7} \approx 0.857$$

Intersection with left boundary.

Left boundary: $x = x_{\min} = 50$

$$y = y_1 + m(x - x_1)$$

$$y = 15 + 0.857(50 - 40)$$

$$y = 23.57$$

Therefore, $A' = (50, 23.57)$

Intersection with top boundary.

Top boundary: $y = y_{\max} = 40$

$$x = x_1 + m(y - y_1)$$

$$x = 40 + 0.857(40 - 15)$$

$$x = 69.17$$

Therefore, $B' = (69.17, 40)$

Final clipped line is $(50, 23.57) \rightarrow (69.17, 40)$

5. Explain polygon surface and B-spline curves.

Defination:

A polygon surface is a 3D object made using many polygon such as triangles and rectangles.

A collection of polygons used to represent a 3D object is called a poly mesh/polygon mesh.

Polygon mesh:

- 1) Explicit representation
- 2) Pointers to a vertex list.
- 3) Pointers to an edge list.

Polygon Tables:

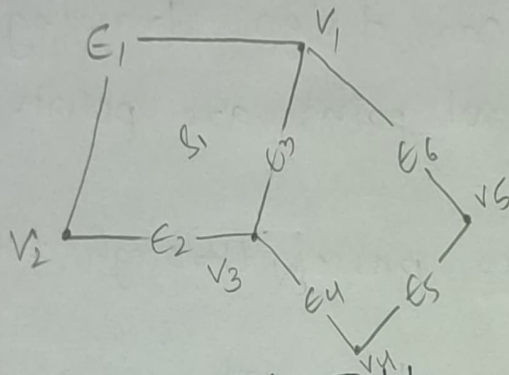
Polygon surfaces can be stored using tables containing.



Vertex table: Stores the x, y, z coordinates of each vertex.

Edge table: Stores the vertices connected by each edge.

Polygon surface table: Stores the edges that makeup each surface.



Vertex Table

$V_1: x_1, y_1, z_1$

$V_2: x_2, y_2, z_2$

$V_3: x_3, y_3, z_3$

$V_4: x_4, y_4, z_4$

$V_5: x_5, y_5, z_5$

Edge Table

$E_1: V_1, V_2$

$E_2: V_2, V_3$

$E_3: V_3, V_4$

$E_4: V_4, V_5$

$E_5: V_5, V_6$

$E_6: V_6, V_1$

Polygon-surface table

$S_1: E_1, E_2, E_3$

$S_2: E_3, E_4, E_5, E_6$

Advantages:

1. Can be used to model almost any object.
2. Easy to represent using vertices.
3. Easy to transform.

Disadvantages

1. Curved surface can only be approximately represented.
2. Objects such as liquid are difficult to simulate using polygon mesh.



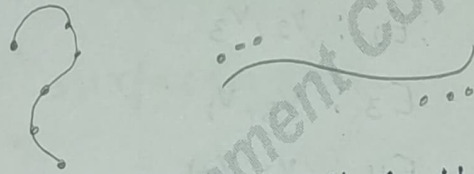
B-Spline curve:

A B-spline curve is a smooth curve used in computer graphics. It is controlled by control points. By moving these points, we can change the shape of curve.

Uses: Animation, car, design, drawings and fonts.

Control point: Control point are points that control shape of the curve.

- Interpolation: Curves passes through the control points.
- Approximation: Curve does not necessarily pass through all control points.



Convex Hull: The convex hull is the boundary that covers control points. The points lie inside or on this boundary.

Characteristic polygon: When the control points are joined with straight lines, the resulting polygon is called the characteristic polygon.

Properties:

- It is a piecewise polynomial curve.
- It can be made using several smaller curve pieces.
- It provides smooth curves.
- Its properties can be local or global.